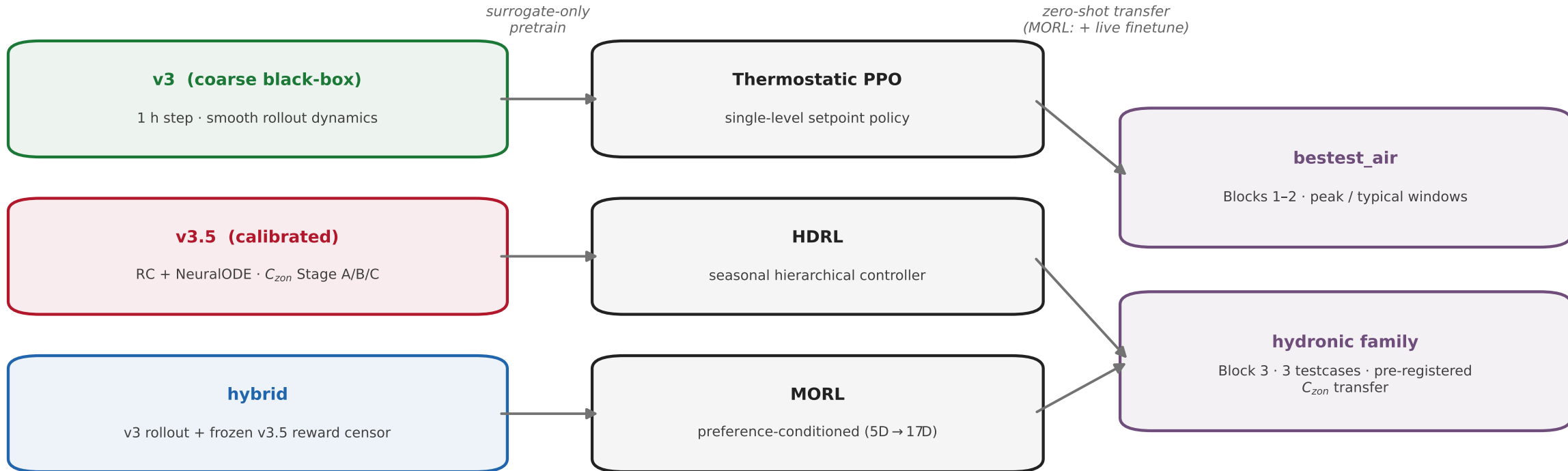


Surrogate training backends

Controller families (PPO-based)

Live BOPTEST evaluation



Evaluation metrics: live maintenance score m_s (= time-in-violation + severity), comfort-violation %, energy (kWh), 24 h rollout $RMSE_T$ · thresholds: $m_s = 1$ collapse, 5% violation, transfer $\tau_k = 1.25 m_s^{Pl}$.

Schematic (no data).